PROJECT SYNOPSIS

MUSHAK ROVER

Autonomous Tracked Unmanned Ground Vehicle with Integrated SHURTHI MK-01 6-DOF Robotic Arm

AUTONOMOUS ROBOTICS • MOBILE MANIPULATION • HUMAN-SUPERVISED AUTONOMY

Complete original synopsis

Abstract

MUSHAK ROVER is an advanced autonomous tracked Unmanned Ground Vehicle (UGV) designed as a multifunctional robotic platform capable of operating in challenging, hazardous, and difficult-to-access environments. The system integrates a tracked mobility platform with autonomous navigation, environmental sensing, computer vision, artificial intelligence, wireless communication, and the SHURTHI MK-01 6-DOF robotic arm into a single intelligent robotic system. The primary objective of MUSHAK ROVER is to reduce human exposure to dangerous environments while extending human capabilities through intelligent robotic assistance. Unlike conventional remotely operated vehicles that primarily depend on continuous human control, the proposed platform is designed to incorporate autonomous perception, localization, path planning, obstacle detection, and navigation while maintaining human supervision over critical decisions. MUSHAK ROVER is envisioned as a modular platform capable of being adapted for defence-support operations, disaster management, search and rescue, industrial inspection, hazardous-environment monitoring, mining, agriculture, infrastructure inspection, and environmental monitoring. The integration of autonomous mobility with the SHURTHI MK-01 robotic arm allows the system not only to reach a target location but also to physically interact with its surroundings and perform controlled tasks. The project aims to demonstrate how robotics, artificial intelligence, sensor fusion, autonomous navigation, and robotic manipulation can be integrated into a scalable next-generation mobile robotic platform.

Introduction

The rapid development of robotics and artificial intelligence has created new possibilities for deploying intelligent machines in environments where human intervention can be dangerous, inefficient, or physically impossible. Unmanned Ground Vehicles have traditionally been developed for specific purposes such as surveillance, transportation, inspection, or remote operation, while robotic arms are generally deployed in fixed industrial environments. MUSHAK ROVER proposes a different approach by integrating autonomous mobility with advanced robotic manipulation through the SHURTHI MK-01 6-DOF robotic arm. The tracked vehicle provides mobility, the sensor suite provides environmental perception, the onboard computing system provides intelligence and decision support, and SHURTHI MK-01 provides physical interaction capability. This integration enables MUSHAK ROVER to travel through an environment, understand its surroundings, reach a designated location, and perform a controlled task while allowing a human operator to supervise the mission.

Problem Statement

Several real-world environments present significant risks to human personnel. Defence and border-support operations may require personnel to inspect difficult terrain or potentially hazardous areas. Disaster zones may contain unstable structures, debris, fire, smoke, or damaged infrastructure. Industrial facilities can contain toxic gases, extreme temperatures, hazardous machinery, and restricted-access zones. Mining environments may involve unstable terrain, poor visibility, dust, and potentially dangerous atmospheric conditions. In many such situations, remotely operated robotic systems can reduce human exposure, but conventional remote-controlled robots still depend heavily on continuous operator input. Furthermore, many robotic platforms are designed around a single function and lack the combination of autonomous mobility, environmental perception, and physical manipulation. MUSHAK ROVER addresses this gap by combining autonomous mobility, intelligent perception, remote supervision, and the SHURTHI MK-01 robotic manipulation system into one modular platform.

Proposed System

MUSHAK ROVER is proposed as a tracked autonomous UGV equipped with the SHURTHI MK-01 6-DOF robotic arm and a multi-sensor perception system. The tracked chassis is intended to provide improved traction and mobility over uneven surfaces such as soil, gravel, debris, and other challenging terrain within the designed operating limits. The platform can incorporate cameras, LiDAR or depth sensors, an IMU, GPS, ultrasonic sensors, and application-specific environmental sensors. These sensors provide information about the robot's surroundings, position, orientation, obstacles, terrain, and environmental conditions. An onboard computing platform processes the collected information and coordinates navigation, perception, communication, and robotic functions. The SHURTHI MK-01 robotic arm provides the system with the ability to perform controlled physical interaction with objects and equipment. The overall system therefore follows a complete operational cycle of sensing the environment, determining its position, planning a route, navigating toward a target, positioning SHURTHI MK-01, performing an assigned task, and communicating the results to a human operator.

Autonomous Navigation and Perception

Autonomous navigation is one of the central components of MUSHAK ROVER. The proposed system can use a combination of LiDAR, cameras, IMU, GPS, and other sensors to estimate its position and understand the surrounding environment. Technologies such as ROS 2, SLAM, Nav2, computer vision, and sensor fusion can be incorporated into the system architecture. SLAM can enable MUSHAK ROVER to simultaneously construct a representation of an unknown environment and estimate its own position within that environment. Once the environment is mapped or a known map is available, the navigation system can generate a path toward a predefined destination while identifying and avoiding obstacles. This enables the rover to perform navigation tasks with reduced dependence on continuous manual control. The autonomous system can also be designed with manual override capabilities so that the human operator can intervene whenever required.

Artificial Intelligence and Computer Vision

Artificial intelligence and computer vision can significantly enhance the perception capabilities of MUSHAK ROVER. Cameras and other sensors can provide visual information that can be processed using computer vision algorithms and AI models. Depending on the application, the system could assist in detecting objects, identifying people, recognizing environmental features, assessing infrastructure damage, classifying terrain, and identifying potential obstacles. In disaster-response applications, computer vision could assist operators in identifying possible survivors, damaged structures, blocked pathways, and other important features. In industrial applications, AI-assisted vision could support equipment inspection and anomaly detection. The objective is not to allow the rover to independently make unrestricted critical decisions, but to provide intelligent perception and decision-support capabilities while maintaining appropriate human supervision.

Defence Applications

MUSHAK ROVER has significant potential as a defence-support robotics platform when configured for non-weaponized applications. In defence environments, the system could be used for remote reconnaissance, perimeter inspection, terrain assessment, infrastructure monitoring, remote observation, logistics assistance, and hazardous-area inspection. The rover could be deployed ahead of personnel to provide visual and environmental information about an area, reducing unnecessary exposure to risk. Its tracked mobility could allow it to operate across terrain that may be difficult for conventional wheeled platforms. The SHURTHI MK-01 robotic arm could provide controlled remote manipulation capabilities for selected non-combat tasks, allowing an operator to interact with equipment or objects from a safer distance. MUSHAK ROVER could therefore serve as a human-support system that improves situational awareness and reduces the need to expose personnel unnecessarily to potentially hazardous environments.

Disaster Management and Search-and-Rescue Applications

The same MUSHAK ROVER platform can be adapted for disaster management and search-and-rescue operations. Earthquakes, landslides, industrial accidents, floods, and structural collapses can create environments that are unsafe for immediate human entry. MUSHAK ROVER could be equipped with thermal cameras, low-light cameras, microphones, speakers, environmental sensors, and gas-detection systems to assist rescue teams. The rover could enter selected areas, transmit live visual information, identify environmental hazards, and assist operators in determining whether an area is safe for human entry. The SHURTHI MK-01 arm could potentially perform lightweight manipulation tasks or interact with specially designed mechanisms. Such capabilities could improve the information available to rescue teams while reducing unnecessary exposure to unstable or hazardous environments.

Industrial and Hazardous-Environment Applications

Industrial environments provide another important application area for MUSHAK ROVER. Factories, chemical facilities, power infrastructure, warehouses, pipelines, and other industrial environments may contain areas where routine human inspection is difficult or hazardous. The rover can be configured with application-specific sensors for temperature, humidity, gas concentration, thermal imaging, or other environmental parameters. It could support remote inspection of equipment, restricted zones, pipelines, infrastructure, and industrial surroundings. For hazardous environments, the platform would require additional engineering such as suitable protective materials, thermal management, environmental sealing, appropriate communication systems, and specialized sensors. Therefore, the student prototype would serve as a technological foundation, while deployment in extreme environments would require extensive application-specific engineering and certification.

Mining and Underground Applications

Mining environments can involve uneven terrain, dust, restricted spaces, low visibility, unstable areas, and potentially hazardous atmospheric conditions. A suitably engineered version of MUSHAK ROVER could support remote inspection, environmental monitoring, mapping, and equipment observation in such environments. Additional gas sensors, thermal cameras, LiDAR systems, and atmospheric monitoring equipment could be integrated depending on operational requirements. The autonomous navigation capability could reduce the need for continuous manual driving and enable systematic inspection of predefined areas.

Agriculture and Environmental Monitoring

Although MUSHAK ROVER is strongly suited to defence-support and hazardous-environment applications, its modular architecture also allows it to be adapted for agriculture and environmental monitoring. In agricultural environments, the platform could be configured for crop observation, environmental sensing, field inspection, and autonomous movement between crop rows. Cameras and sensors could collect information about crop conditions and environmental parameters. Similarly, environmental-monitoring configurations could allow the rover to collect temperature, humidity, air-quality, gas, soil, and other application-specific measurements at different locations. This demonstrates the potential for MUSHAK ROVER to serve multiple sectors without requiring a completely new robotic system for every application.

SHURTHI MK-01 6-DOF Robotic Arm

The SHURTHI MK-01 is the dedicated 6-DOF robotic manipulation system mounted on MUSHAK ROVER. A conventional mobile robot may be capable of reaching a particular location but have limited ability to interact with its surroundings. SHURTHI MK-01 provides the additional capability of physical manipulation. Six degrees of freedom allow the arm to achieve a wider range of positions and orientations, making it suitable for controlled interaction with selected objects and equipment. Depending on the end-effector design, SHURTHI MK-01 could be used for inspection, picking and placing lightweight objects, interacting with specially designed mechanisms, sample collection, or other predefined tasks. The arm can also be designed around interchangeable end-effectors, allowing the same MUSHAK ROVER platform to be adapted for different missions.

Human-Supervised Autonomy

A fundamental principle of MUSHAK ROVER is human-supervised autonomy. The purpose of autonomy is to reduce repetitive manual control and improve operational efficiency rather than remove humans from important decision-making processes. MUSHAK ROVER can autonomously perform tasks such as localization, mapping, obstacle detection, path planning, and navigation while the human operator retains mission-level supervision and the ability to intervene. A communication interface can provide live video, rover location, battery status, sensor information, SHURTHI MK-01 arm status, mission progress, and alerts. Manual override and emergency-stop systems can be included to ensure that the operator can immediately regain control whenever necessary. This human-in-the-loop architecture is particularly important when the rover is considered for sensitive environments.

Communication and Monitoring System

MUSHAK ROVER can incorporate a dedicated monitoring interface through which an operator can observe and control the system. The interface can display live camera feeds, rover position, battery status, sensor readings, navigation information, robotic-arm status, and system alerts. Depending on the deployment environment, communication technologies such as Wi-Fi, long-range radio, or cellular networks could be considered. For future deployment in defence or critical infrastructure environments, secure authentication, encryption, communication redundancy, and cybersecurity mechanisms would be essential. The communication architecture can therefore be developed as a modular component that can be adapted according to mission requirements.

Safety Architecture

Safety is an important part of the proposed MUSHAK ROVER system because autonomous robots may operate in unpredictable environments. The rover can incorporate emergency-stop mechanisms, manual override, obstacle avoidance, speed limitations, geofencing, motor shutdown procedures, battery monitoring, communication-loss failsafes, SHURTHI MK-01 movement limits, and system-health monitoring. If the rover detects a critical system condition or loses communication with the operator, it can be programmed to transition into a predefined safe state. These mechanisms are intended to ensure that autonomy remains controlled and predictable during testing and operation.

Modularity and Multi-Domain Design

One of the key design philosophies of MUSHAK ROVER is modularity. The same tracked platform can be configured with different sensors and payloads depending on the intended mission. A defence-support configuration may prioritize cameras, LiDAR, GPS, communication systems, and environmental monitoring. A search-and-rescue configuration could incorporate thermal imaging, microphones, speakers, and gas sensors. An industrial configuration could use thermal and equipment-inspection sensors, while an agricultural configuration could include crop-monitoring cameras and environmental sensors. The SHURTHI MK-01 arm can similarly support interchangeable end-effectors. This modular approach allows MUSHAK ROVER to evolve without requiring a complete redesign for every application.

Proposed Technology

The hardware architecture of MUSHAK ROVER can consist of a tracked chassis, DC geared motors, motor controllers, an embedded microcontroller, an onboard computer such as a Raspberry Pi or Jetson-class system, LiDAR or depth sensors, cameras, an IMU, GPS, ultrasonic sensors, environmental sensors, the SHURTHI MK-01 6-DOF robotic arm, battery-management components, and wireless communication hardware. The software architecture can be based on Linux, ROS 2, Python, C/C++, OpenCV, SLAM, Nav2, AI-based computer-vision models, and a dedicated monitoring dashboard. The exact hardware can be scaled according to the available budget and prototype requirements.

Development Methodology

The development of MUSHAK ROVER can be divided into multiple stages. The first stage involves mechanical development of the tracked chassis, motor system, battery compartment, and SHURTHI MK-01 mounting structure. The second stage focuses on embedded motor control, power management, emergency-stop mechanisms, and basic mobility. The third stage involves integrating sensors such as cameras, LiDAR, IMU, GPS, and environmental sensors. The fourth stage focuses on autonomous navigation, localization, mapping, path planning, and obstacle avoidance. The fifth stage involves integrating SHURTHI MK-01 and developing coordinated control between the mobile platform and robotic manipulator. The sixth stage introduces AI-assisted computer vision and environmental perception. The seventh stage develops the communication and monitoring interface. Finally, the complete system can be tested under controlled conditions involving uneven terrain, obstacles, navigation routes, remote operation, robotic manipulation, communication reliability, and sensor performance.

Expected Outcome

The expected outcome of the project is a functional prototype of MUSHAK ROVER capable of demonstrating autonomous or semi-autonomous tracked mobility, environmental perception, obstacle detection, navigation, remote monitoring, and controlled operation of the SHURTHI MK-01 6-DOF robotic arm. The prototype should demonstrate that a relatively compact robotic platform can integrate multiple advanced technologies into one system. Performance can be evaluated through measurable parameters such as navigation accuracy, obstacle-detection performance, route-completion rate, localization accuracy, communication reliability, battery endurance, terrain capability, robotic-arm positioning accuracy, and manipulation success rate. These measurements can provide quantitative evidence of the system's effectiveness and establish a foundation for future improvements.

Innovation and Research Significance

The novelty of MUSHAK ROVER does not depend on claiming that tracked UGVs, robotic arms, LiDAR, AI, or autonomous navigation are individually new technologies. Instead, the research significance lies in their integration into a modular mobile-manipulation platform designed around multi-domain applications. The combination of tracked mobility, autonomous navigation, sensor fusion, AI-assisted perception, the SHURTHI MK-01 robotic arm, human supervision, and interchangeable mission payloads creates a platform that can be adapted for different operational requirements. This integrated architecture provides a strong foundation for further research in autonomous robotics, mobile manipulation, human-robot collaboration, and intelligent field robotics.

Future Scope

Future versions of MUSHAK ROVER could incorporate advanced LiDAR, stereo or depth vision, thermal imaging, edge AI processing, improved SLAM, autonomous docking, automated battery charging, terrain classification, advanced SHURTHI MK-01 coordination, multi-robot collaboration, improved communication systems, specialized industrial sensors, digital-twin simulation, and fleet-management capabilities. A larger version could be developed to carry heavier payloads, while smaller versions could be optimized for inspection and reconnaissance. The platform could also evolve toward more sophisticated mobile-manipulation research in which MUSHAK ROVER autonomously determines how to position itself before performing a manipulation task using SHURTHI MK-01.

Conclusion

MUSHAK ROVER, integrated with the SHURTHI MK-01 6-DOF robotic arm, represents a transition from conventional remote-controlled robotic vehicles toward intelligent autonomous mobile robotic systems capable of sensing, navigating, assisting, and physically interacting with their surroundings. By integrating tracked mobility, autonomous navigation, artificial intelligence, computer vision, multi-sensor perception, wireless communication, and robotic manipulation, the project establishes a versatile platform with potential applications in defence support, disaster management, search and rescue, industrial inspection, mining, agriculture, infrastructure monitoring, and hazardous-environment assessment. The system is designed around the principle that robots should extend human capabilities rather than replace human judgment in critical situations. Its modular architecture allows the platform to evolve according to application requirements and provides a strong foundation for future research and development. MUSHAK ROVER therefore aims not merely to demonstrate a robotic vehicle, but to explore the development of an intelligent, adaptable, and human-supervised robotic platform capable of supporting people in environments where conventional approaches may be unsafe, inefficient, or impractical.